Unit 7: First-Order Exactness and Methods

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Condensed Cheat Sheet

Unit 7 at a Glance: Core Sub-Topics

A first-order ODE written as $M\,dx + N\,dy = 0$ is *exact* when its left side is already the total differential dF of some scalar potential F, making the general solution simply $F(x, y) = C$. This unit develops the complete toolkit: test the cross-partials to detect exactness, reconstruct the potential via the integration loop, repair non-exact equations with special integrating factors, connect the theory to conservative vector fields, and read the physics through thermodynamic state functions.

- **7.1 Exact Differential Equations** — the criterion $M_y = N_x$ and the implicit general solution $F(x, y) = C$.
- **7.2 Exact Equations Intuition** — Clairaut's theorem: $F_{xy} = F_{yx}$ is why $M_y = N_x$ guarantees a potential.
- **7.3 Exact Equations Motivation and Theory** — the integration loop: $F = \int M\,dx + g(y)$, fix $g'(y)$ by matching $F_y = N$.
- **7.4 Non-Exact Equations and Special Integrating Factors** — integrating factor $\mu(x)$ when Δ/N is pure in x; $\mu(y)$ when $-\Delta/M$ is pure in y (where $\Delta = M_y - N_x$).
- **7.5 Almost Exact Differential Equations** — decision tree: test both $\mu(x)$ and $\mu(y)$ routes systematically, then execute the repair and re-verify exactness before reconstructing F.
- **7.6 Exact Equations and Conservative Vector Fields** — $M_y = N_x$ iff $\mathbf{F} = \langle M, N\rangle$ is conservative; path independence via the Fundamental Theorem for Line Integrals.
- **7.7 Applications in Thermodynamics** — dU is an exact state differential; δQ is inexact; the integrating factor $1/T$ produces exact entropy $dS = \delta Q/T$.

Part I — Condensed Cheat Sheet

7.1 Exact Differential Equations

A first-order ODE $M(x, y)\,dx + N(x, y)\,dy = 0$ is *exact* on a simply connected region when its left side equals the total differential dF of some potential function $F(x, y)$. The key theorem reduces exactness to a single algebraic test on the cross-partial derivatives of M and N.

Exactness Criterion and Implicit Solution

The ODE $M\,dx + N\,dy = 0$ (with M, N having continuous first partials) is **exact** on a simply connected region if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

When exact, a potential $F(x, y)$ exists satisfying $F_x = M$ and $F_y = N$, and the **general solution** is

$$F(x, y) = C.$$

Common Pitfall: Swapped Cross Partials

Always differentiate M with respect to y and N with respect to x. The test is $M_y = N_x$, never M_x versus N_y. Computing the same-variable partials tells you nothing about exactness and will produce a false positive or false negative on every equation you test.

7.2 Exact Equations Intuition

The cross-partial criterion is not an arbitrary rule — it is Clairaut's theorem in disguise. If $M = F_x$ and $N = F_y$ for some F, then $M_y = F_{xy}$ and $N_x = F_{yx}$, and Clairaut guarantees the two mixed partials agree whenever F has continuous second partials. Exactness is precisely the statement that M and N are partials of a common function.

Clairaut's Theorem and the Exact-Differential Connection

On a simply connected region with M, N continuously differentiable,

$$M\,dx + N\,dy = dF \iff \frac{\partial M}{\partial y} = \frac{\partial^2 F}{\partial y\,\partial x} = \frac{\partial^2 F}{\partial x\,\partial y} = \frac{\partial N}{\partial x}.$$

The equality of mixed partials is both **necessary** (if dF exists then the cross partials must agree) and **sufficient** (on a simply connected domain, equal cross partials guarantee a global potential).

7.3 Exact Equations Motivation and Theory

Once exactness is confirmed, the potential F is found by a two-step integration loop. Integrating M with respect to x introduces an arbitrary function $g(y)$ in place of a constant; that function is then pinned down by the condition $F_y = N$.

Potential Function Reconstruction: The Integration Loop

Given $M\,dx + N\,dy = 0$ confirmed exact by $M_y = N_x$:

1. **Integrate M in x**, holding y fixed:

$$F(x, y) = \int M(x, y)\,dx + g(y).$$

2. **Differentiate F in y** and set the result equal to N:

$$\frac{\partial F}{\partial y} = N(x, y) \quad \Longrightarrow \quad g'(y) = N(x, y) - \frac{\partial}{\partial y}\int M\,dx.$$

   (All x-terms must cancel; if they do not, re-check the exactness test.)

3. **Integrate** $g'(y)$ in y to obtain $g(y)$.
4. Write the **general solution** $F(x, y) = C$.

Common Pitfall: $g'(y)$ Must Be Free of x

After forming $g'(y) = N - \partial_y \int M\,dx$, every x-term must cancel. If x survives in $g'(y)$, the equation was not exact (or an arithmetic error occurred). Integrating an x-dependent expression as if it were a function of y alone produces a wrong answer without any LaTeX or algebra error to signal the problem.

7.4 Non-Exact Equations and Special Integrating Factors

When $M_y \neq N_x$, no potential F exists for the equation as written. Multiplying both sides by a non-zero *integrating factor* μ can change the cross partials of μM and μN until they match. Two special forms of μ are algebraically testable and cover the majority of non-exact equations encountered in practice.

Integrating Factors $\mu(x)$ and $\mu(y)$

Let $\Delta = M_y - N_x$ (the **exactness defect**).

- **Test for $\mu(x)$:** Compute $\dfrac{\Delta}{N}$. If the result depends on x *only*, set $h(x) = \Delta/N$ and

$$\mu(x) = \exp\left(\int h(x)\,dx\right).$$

- **Test for $\mu(y)$:** Compute $\dfrac{N_x - M_y}{M} = \dfrac{-\Delta}{M}$. If the result depends on y *only*, set $k(y) = -\Delta/M$ and

$$\mu(y) = \exp\left(\int k(y)\,dy\right).$$

After multiplying the ODE through by μ, solve the resulting exact equation via the integration loop (Section 7.3).

Common Pitfall: Always Re-Verify Exactness After Multiplying

The formulas for $\mu(x)$ and $\mu(y)$ are derived by assuming the repaired equation will be exact. After multiplying by μ, explicitly re-compute the cross partials $(\mu M)_y$ and $(\mu N)_x$ and confirm they are equal before proceeding. A mis-simplified defect ratio can give a wrong μ that does *not* restore exactness.

7.5 Almost Exact Differential Equations

An *almost exact* equation is non-exact but possesses an integrating factor of one of the two standard forms ($\mu(x)$ or $\mu(y)$). The key skill is recognizing which test succeeds and executing the full repair-and-solve workflow systematically.

Integrating-Factor Decision Tree

Given $M\,dx + N\,dy = 0$ with $M_y \neq N_x$:

1. Compute the defect $\Delta = M_y - N_x$.
2. **Try** $\mu(x)$**:** Form Δ/N. Does all y cancel? If yes, $\mu(x) = e^{\int \Delta/N\,dx}$.
3. **Try** $\mu(y)$**:** Form $-\Delta/M$. Does all x cancel? If yes, $\mu(y) = e^{\int(-\Delta/M)\,dy}$.
4. Multiply the ODE by μ, re-verify $(\mu M)_y = (\mu N)_x$, then use the integration loop to find F and write $F = C$.

Why the Integrating Factor ODE Works

Requiring $(\mu M)_y = (\mu N)_x$ for $\mu = \mu(x)$ yields

$$\mu\,M_y = \mu'\,N + \mu\,N_x \quad\Longrightarrow\quad \frac{\mu'}{\mu} = \frac{M_y - N_x}{N} = \frac{\Delta}{N},$$

a separable ODE in $\mu(x)$ whose solution is $\mu = e^{\int \Delta/N\,dx}$. The analogous calculation with $\mu = \mu(y)$ gives $\mu'/\mu = -\Delta/M$.

7.6 Exact Equations and Conservative Vector Fields

Every exact equation $M\,dx + N\,dy = 0$ can be read as the zero-work condition $\mathbf{F}\cdot d\mathbf{r} = 0$ for the vector field $\mathbf{F} = \langle M, N\rangle$. Exactness is equivalent to $\mathbf{F}$ being *conservative*, and the potential F of the ODE is the scalar potential φ with $\nabla\varphi = \mathbf{F}$.

Conservative Fields, Scalar Potentials, and Path Independence

A C^1 vector field $\mathbf{F} = \langle M, N\rangle$ on a simply connected region is **conservative** if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

When conservative, there exists a scalar potential φ with $\nabla\varphi = \mathbf{F}$, found by the same integration loop as the ODE potential. Line integrals are then **path independent**:

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \varphi(B) - \varphi(A)$$

for any smooth curve C from point A to point B (Fundamental Theorem for Line Integrals). Equivalently, $\oint_C \mathbf{F}\cdot d\mathbf{r} = 0$ for every closed curve.

Common Pitfall: The Domain Must Be Simply Connected

The equivalence $M_y = N_x \iff$ conservative holds only on a **simply connected** domain (a domain with no holes). On a multiply-connected domain such as $\mathbb{R}^2 \setminus \{(0,0)\}$, the field $\mathbf{F} = \langle -y,\, x\rangle/(x^2+y^2)$ satisfies $M_y = N_x$ everywhere it is defined yet is *not* conservative: its integral around the unit circle equals $2\pi \neq 0$.

Unit 7 Key Takeaway

Every exact equation $M\,dx + N\,dy = 0$ is secretly $dF = 0$; its general solution is $F(x,y) = C$ recovered by the integration loop $F = \int M\,dx + g(y)$ with $g'(y)$ fixed by matching $F_y = N$. When $M_y \neq N_x$, form the defect $\Delta = M_y - N_x$ and repair with $\mu(x) = e^{\int \Delta/N\,dx}$ if Δ/N is pure in x, or $\mu(y) = e^{-\int \Delta/M\,dy}$ if $-\Delta/M$ is pure in y. Exactness is conservatism: $M_y = N_x \iff \mathbf{F} = \nabla\varphi$, giving path independence $\int_C \mathbf{F}\cdot d\mathbf{r} = \varphi(B) - \varphi(A)$, and the integrating factor $1/T$ converts inexact thermodynamic heat δQ into the exact entropy differential dS.